

1. **Execute the basic and user status commands like: su, sudo, man, help, history, who, whoami, id, uname, uptime, free, tty, cal, date, hostname, reboot, clear and their options**

Objective

To introduce students to Linux system administration, user session management, privilege escalation, system health monitoring, and terminal control commands, developing fundamental skills for auditing user activity and inspecting system resources.

Learning Outcome:

By the end of this lab, students will be able to:

- **Verify** active user contexts, effective user IDs, group memberships, and terminal sessions using whoami, id, who, and tty.
- **Elevate and switch** user privileges securely using su and sudo.
- **Monitor** system status, kernel specifications, resource utilization, uptime, and system clock details using uname, uptime, free, hostname, date, and cal.
- **Retrieve** built-in documentation and command execution history using man, help, and history.
- **Manage** terminal displays and execute controlled system restarts using clear and reboot.

Procedure:

i) User Identity & Session Management

whoami

Displays the effective username of the current user session.

whoami

id

Prints user and group information for the specified user (or current user if omitted).

id [OPTIONS] [USERNAME]

- **-u:** Prints only the numeric User ID (UID).
- **-g:** Prints only the primary Group ID (GID).

- **-G**: Prints all supplementary Group IDs.
- **-n**: Displays names instead of numeric IDs (used with -u, -g, or -G).

Example: id -un (Prints the current username)

who

Shows who is logged on to the system. This shows, for each logged-in user: **username, terminal (tty/pts), login date and time, and sometimes the remote host they connected from**

who [OPTIONS]

- **-H (--heading)**: Prints column headers (NAME, LINE, TIME, COMMENT).
- **-b**: Displays the time of the last system boot.
- **-a (--all)**: Displays all detailed session information available.

tty

The most basic use of the tty command is to display the name of the current terminal. Prints the file name of the terminal connected to standard input (e.g., /dev/pts/0).

/dev/pts/0

| | |

| | └─ 0 = terminal number

| └── pts = pseudo-terminal

└── dev = device

tty [OPTIONS]

- **-s (--silent)**: Silent mode; prints nothing, returning exit status only.

ii). Privilege Escalation Commands

su (Switch User)

The su command in Linux is used to switch to another user account (or to the root/superuser) from within a current session. Switches the current shell session to another user context.

su [OPTIONS] [USER]

Create a second regular user (more realistic practice)

```
sudo adduser testuser
```

It'll prompt you to set a password and some optional details (press Enter to skip them). Then:

```
su testuser
```

Enter testuser's password, and you'll switch to that account.

Common use cases:

1. Become root for a session:

```
su -
```

Prompts for the root password, then drops you into a root shell where every command runs with full privileges — useful when you need to run many admin commands in a row (rather than prefixing each one with sudo).

2. Switch to a specific user:

```
su - username
```

Logs you in as that user (needs their password), e.g. to test something under a service account.

3. Run a single command as another user without a full shell switch:

```
su -c "command" username
```

press **Ctrl+d** to exit or type **"exit"** in terminal

su vs sudo — the key difference:

- su asks for the target user's password (e.g. root's password) and switches your whole session to that user until you exit.
- sudo asks for your own password and runs a single command with elevated privileges, then returns you to your normal user — no full switch happens.

sudo (Superuser Do)

Runs command as root, prompting for your own password. Executes a command with superuser (root) or alternative account privileges as defined in /etc/sudoers.

Bash

sudo [OPTIONS] COMMAND

Option	Meaning	Example
-u	Run the command as a specific user instead of root	sudo -u testuser whoami
-i	Start an interactive root login shell (loads root's own environment/profile, like logging in as root)	sudo -i
-l	List the commands your account is allowed to run with Sudo	sudo -l
-v	Validate/refresh your sudo timestamp — extends your cached sudo session without running a command	sudo -v
-k	Invalidate (kill) your cached sudo timestamp — next sudo command will ask for your password again right away	sudo -k

Note : press **Ctrl+d** to exit the session.

iii). System Telemetry & Environment Commands

uname (Unix Name)

Prints system and kernel information.

uname [OPTIONS]

- **-a (--all)**: Prints all system information in sequence.
- **-s**: Displays kernel name (e.g., Linux).
- **-r**: Displays kernel release version.
- **-m**: Displays hardware machine architecture (e.g., x86_64, aarch64).

uptime

Tells how long the system has been running, how many users are logged in, and system load averages over 1, 5, and 15 minutes.

uptime [OPTIONS]

- **-p (--pretty)**: Displays system uptime in a human-readable format.
- **-s (--since)**: Displays the date and time the system has been running since.

free

Displays the total amount of free and used physical memory (RAM) and swap space in the system.

free [OPTIONS]

- **-h (--human)**: Displays output in human-readable units (e.g., 1K, 234M, 2G).
- **-m**: Displays memory quantities in megabytes.
- **-g**: Displays memory quantities in gigabytes.
- **-s <seconds>**: Continuously updates memory stats at specified second intervals.

Note : **Gi** stands for **Gibibyte** (short for **giga binary byte**). **Mebibytes (Mi)**

hostname

Displays or sets the system's network name.

hostname [OPTIONS]

- **-i**: Displays network address(es) bound to the hostname.
- **-l**: Displays all network addresses configured on all interfaces.
- **-f**: Displays the Fully Qualified Domain Name (FQDN).

date

Displays or sets the system date and time.

date [OPTIONS] [+FORMAT]

- **-u (--utc)**: Displays or sets time in Coordinated Universal Time (UTC).
- **+"%Y-%m-%d %H:%M:%S"**: Formats output using custom time syntax.
- **-s "<string>"**: Sets system date and time (requires root access).

cal (Calendar)

Displays a simple calendar in the terminal window.

cal [OPTIONS] [[MONTH] YEAR]

- **-3**: Shows the previous, current, and next month side-by-side.
- **-y**: Displays the complete calendar for the current year.
- **-j**: Displays Julian dates (days numbered 1–365).

iv). Help & Navigation Aids

man (Manual Pages)

Interface used to view reference manuals for system commands, system calls, and library functions.

man [SECTION] COMMAND

- **-k <keyword> (apropos)**: Searches manual descriptions for matching keywords.
- **-f <command> (whatis)**: Displays short single-line manual page summaries.

help

Displays information about Bash shell built-in commands (like cd, alias, echo).

help [OPTIONS] [BUILTIN_COMMAND]

- **-d**: Displays a short description for each topic.
- **-m**: Displays pseudo-manpage output format.

history

Prints or manipulates the shell command history list.

history [OPTIONS] [N]

- **N**: Prints only the last N executed lines.
- **-c**: Clears the current shell history list.
- **-d <offset>**: Deletes the entry at the specified history line position.

v). Terminal Control & Power Management

clear

Clears the terminal screen display area.

clear

- **-x**: Clears the viewable screen buffer while preserving the terminal scrollback history.

reboot

Restarts the system operating system immediately.

reboot [OPTIONS]

- **-f (--force)**: Forces immediate reboot without invoking system shutdown services.
- **-p (--poweroff)**: Powers off the machine instead of restarting.